Exercise 1

Conditions:

① $Z_a = 70 + j30\,\Omega$

① $Z_a = Z_0 = 50\,\Omega$

$V_g = 10V$

$V_a = 2V$

$Z_g = 30 + j20\,\Omega$

$Z_{RX} = 60 - 25j\,\Omega$

$Z_{Ri} = Z_0 = 50\,\Omega$ for $i = 1,2,3$

$$S = \begin{pmatrix} 0 & e^{j\varphi_1} & 0 \\ 0 & 0 & e^{j\varphi_2} \\ e^{j\varphi_3} & 0 & 0 \end{pmatrix} \text{ with } \varphi_1 = \varphi_2 = \varphi_3 = \tfrac{2\pi}{3}$$

Configuration a - TX operation, we evaluate P_3/P_2

Power enters at port ② and it is redirected to port ①:

$b_1 = S_{12}\, a_2$

At port ① part of the signal is reflected back:

$$\Gamma_a = \frac{Z_a - Z_0}{Z_a + Z_0} \longrightarrow a_1 = \Gamma_a b_1 = \Gamma_a S_{12}\, a_2$$

The reflected signal then goes back into port ① and feeds into port ③

$b_3 = S_{31}\, a_1 = S_{31}\, \Gamma_a b_1 = S_{31}\, \Gamma_a S_{12}\, a_2$

At port ③ the signal is partially reflected into port ②

$a_3 = \Gamma_{RX}\, b_3 \longrightarrow b_2 = S_{23}\, a_3 = S_{23}\, \Gamma_{RX}\, S_{31}\, \Gamma_a S_{12}\, a_2$

Since the power delivered at port ② is:

$P_2 = |a_2|^2 - |b_2|^2 = |a_2|^2 - |\cancel{S_{23} S_{31} S_{12}}|^2 |\Gamma_{RX} \Gamma_a a_2|^2 = |a_2|^2 \left(1 - |\Gamma_{RX} \Gamma_a|^2\right)$

The power absorbed at port 3, by the receiver, like passive dissipation is:

$P_3 = |b_3|^2 \left(1 - |\Gamma_{RX}|^2\right) = |\cancel{S_{31} S_{12}}|^2 |\Gamma_a a_2|^2 \left(1 - |\Gamma_{RX}|^2\right) \Rightarrow |\Gamma_a|^2 |a_2|^2 \left(1 - |\Gamma_{RX}|^2\right)$

So:

$$\frac{P_3}{P_2} = \frac{|\Gamma_a|^2 \left(1 - |\Gamma_{RX}|^2\right)}{1 - |\Gamma_a \Gamma_{RX}|^2}$$

$$\Gamma_{RX} = \frac{Z_{RX} - Z_0}{Z_{RX} + Z_0} = \frac{60 - j25 - 50}{60 - j25 + 50} = \frac{10 - j25}{110 - j25} = 0{,}136 - 0{,}196j \Rightarrow |\Gamma_{RX}|^2 = 0{,}0569$$

- For condition ①

$$\Gamma_a = \frac{Z_a - Z_0}{Z_a + Z_0} = \frac{70 + j30 - 50}{70 + j30 + 50} = 0{,}216 + 0{,}136j \Rightarrow |\Gamma_a|^2 \approx 0{,}0849$$

$$\frac{P_3}{P_2} = \frac{0{,}0849\,(1-0{,}0569)}{1-|(0{,}216+0{,}196j)(0{,}136-0{,}196j)|^2} = 0{,}0803$$

- For condition ②

$Z_a = Z_0 = 50\,\Omega \Rightarrow \Gamma_a = 0 \Rightarrow \frac{P_3}{P_2} = 0 \rightarrow$ There is no reflection on the antenna because its impedence is matched, so there are no parassitic losses!

Configuration b - RX Operation, we evaluate P_3/P_1

In this mode the signal originates at port ①

Power enters at port ① and flows to port ③:

$b_3 = S_{31}\, a_1$

At port 3 a part is reflected

$a_3 = \Gamma_{RX}\, b_3 = \Gamma_{RX}\, S_{31}\, a_1$

It travels into port ②

$b_2 = S_{23}\, a_3 = S_{23}\, \Gamma_{RX}\, S_{31}\, a_1$

At port ②, since the TX circuit acts as a load, it reflects:

$a_2 = \Gamma_g\, b_2 = \Gamma_g\, S_{23}\, \Gamma_{RX}\, S_{31}\, a_1$

And it returns back into port ①

$b_1 = S_{12}\, a_2 = S_{12}\, \Gamma_g\, S_{23}\, \Gamma_{RX}\, S_{31}\, a_1 \rightarrow S_{12} = S_{23} = S_{31} = e^{j\pi/3} \Rightarrow$

$\Rightarrow S_{12} S_{23} S_{31} = \left(e^{j\pi/3}\right)^3 = e^{j\pi} = -1 \Rightarrow$

$\Rightarrow b_1 = -\Gamma_g\, \Gamma_{RX}\, a_1$

Since:

$P_1 = |a_1|^2 - |b_1|^2 = |a_1|^2 - |-\Gamma_g \Gamma_{RX}\, a_1|^2 = |a_1|^2\left(1 - |-\Gamma_g \Gamma_{RX}|^2\right) = |a_1|^2\left(1 - |\Gamma_g \Gamma_{RX}|^2\right)$

→ we don't need this sign, this is surely positive

And:

$P_3 = |b_3|^2\left(1 - |\Gamma_{RX}|^2\right) = |S_{31}\, a_1|^2\left(1 - |\Gamma_{RX}|^2\right) = \cancel{|S_{31}|^2}^{\,1}\, |a_1|^2\left(1 - |\Gamma_{RX}|^2\right)$

the ratio becomes:

$$\frac{P_3}{P_1} = \frac{1 - |\Gamma_{RX}|^2}{1 - |\Gamma_g \Gamma_{RX}|^2}$$

→ This is independent from Z_0, so we have just one single answer for both conditions!

$$\Gamma_g = \frac{Z_g - Z_0}{Z_g + Z_0} = \frac{30 + j20 - 50}{30 + j20 + 50} = -0{,}176 + 0{,}294j \Rightarrow |\Gamma_g|^2 \approx 0{,}118$$

$$\frac{P_3}{P_1} = \frac{1 - 0{,}118}{1 - |(-0{,}176 + 0{,}294j)(0{,}136 - 0{,}196j)|^2} \approx 0{,}949$$

Exercise 2

$$S = \begin{pmatrix} 0 & S_{12}^{circ} & 0 \\ 0 & 0 & S_{23}^{circ} \\ S_{31}^{circ} & 0 & 0 \end{pmatrix} \qquad \frac{V_2}{V_1} = A e^{j\alpha}$$

$$|S_{12}^{circ}| = |S_{23}^{circ}| = |S_{31}^{circ}| = 1$$

$$\angle S_{12}^{circ} = \angle S_{23}^{circ} = \angle S_{31}^{circ} = \pi/3$$

$Z_0 = 50\,\Omega$

$A = 0{,}5$

$\alpha = \pi$

This is a counter-clockwise (RH) circulator

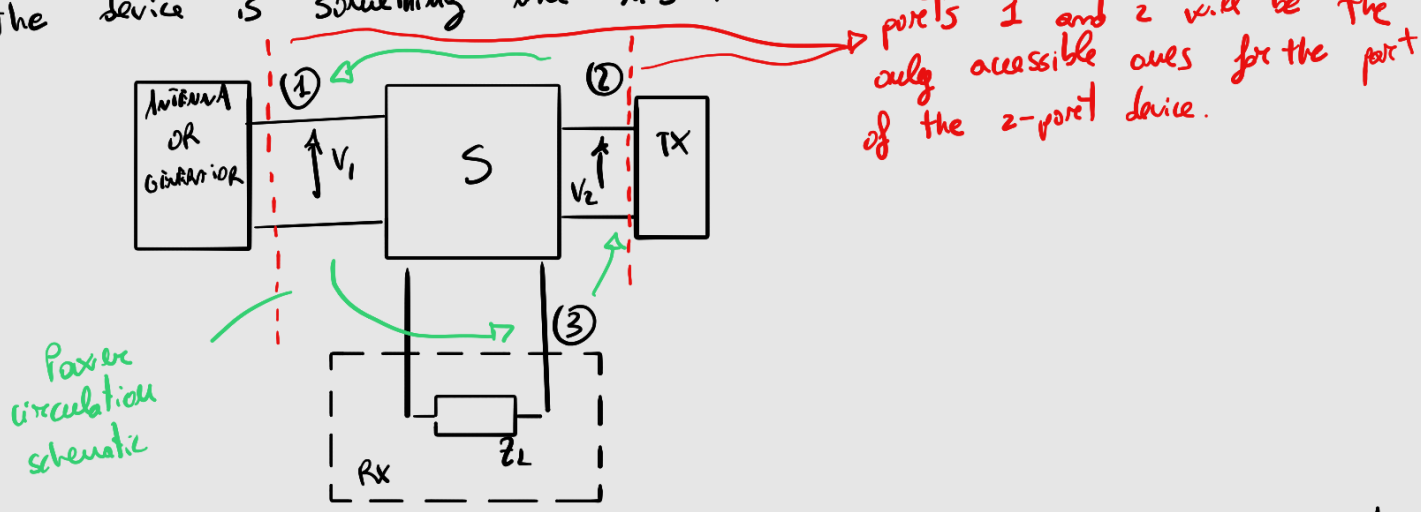
We can determine the power flow

$b_1 = S_{12}^{circ}\, a_2$ (power from port ② to port ①)

$b_2 = S_{23}^{circ}\, a_3$ (power from port ③ to port ②)

$b_3 = S_{31}^{circ}\, a_1$ (power from port ① to port ③)

The device is something like this:



Now for the analytical expression of $\underline{\underline{S}}$, by closing port 3 on a load we can write that:

$a_3 = \Gamma b_3$

And with this and the original circulator matrix we can write the new 2-port device's parameters.

$$S_{11} = \frac{b_1}{a_1}\Big|_{a_2=0} = 0$$

$$S_{21} = \frac{b_2}{a_1}\Big|_{a_2=0} = S_{23}^{circ}\,\Gamma'\,S_{31}^{circ}$$

$$S_{12} = \frac{b_1}{a_2}\Big|_{a_1=0} = S_{12}^{circ}$$

$$S_{22} = \frac{b_2}{a_2}\Big|_{a_1=0} = 0$$

The new scattering matrix for this device is:

$$S = \begin{pmatrix} 0 & S_{23}^{circ}\,\Gamma'\,S_{31}^{circ} \\ S_{12}^{circ} & 0 \end{pmatrix}$$

If $S_{12}^{circ} = S_{23}^{circ}\,\Gamma'\,S_{31}^{circ}$ then this new device is reciprocal, in order for it to be reciprocal it should be:

$$\Gamma' = \frac{S_{12}^{circ}}{S_{23}^{circ}\,S_{31}^{circ}} = \frac{e^{j\pi/3}}{e^{j\pi/3}\,e^{j\pi/3}} = \frac{e^{j\pi/3}}{e^{j\frac{2}{3}\pi}} = e^{j\pi/3}\,e^{-j\frac{2}{3}\pi} = e^{j\frac{\pi}{3}(1-2)} = e^{-j\frac{\pi}{3}} = 0{,}5 - 0{,}866j$$

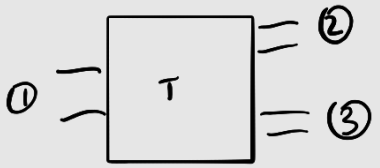
Value needed in order to be reciprocal

For the load's reflection coefficient we know that in nominal condition the voltage ratio is equal to the forwarded transmission coefficient:

$$\frac{V_2}{V_1} = S_{21} = S_{23}^{circ}\,\Gamma'\,S_{31}^{circ} \Rightarrow 0{,}5\,e^{j\pi} = \Gamma' e^{j\frac{2\pi}{3}} \Rightarrow \Gamma' = 0{,}5\,e^{j\pi}\,e^{-j\frac{2\pi}{3}} = 0{,}5\,e^{j\pi/3}$$

So with a load that grant $|\Gamma'| = 0{,}5$ and $\angle\Gamma' = \pi/3$ we can obtain that expression for V_2/V_1

Exercise 3



- port 1 matched $S_{11} = 0$
- output power ratio $\frac{P_3}{P_2} = a = 1{,}25$

$Z_{R1} = Z_{R2} = Z_{R3} = 50\,\Omega$

$Z_{\infty 2}$ and $Z_{\infty 3}$ $\lambda/4$ lenght (90° electrical lenght)

$\varepsilon_r = 4{,}45$ and $h = 1{,}55$ mm

$f_0 = 5{,}8$ GHz $\Rightarrow \lambda = \frac{c}{f_0} = 51{,}72$ mm

The first part of the exercise is to get all the values of impedance for each line so that we know how to get w and l of the microstrips.

Port 1 is matched $Z_{0l} = 50\,\Omega$

Now we proceed with the T-junction:

$$\frac{P_3}{P_2} = \frac{Z_{\infty 2}^2}{Z_{\infty 3}^2} \Rightarrow Z_{\infty 2}^2 = 1{,}25\, Z_{\infty 3}^2$$

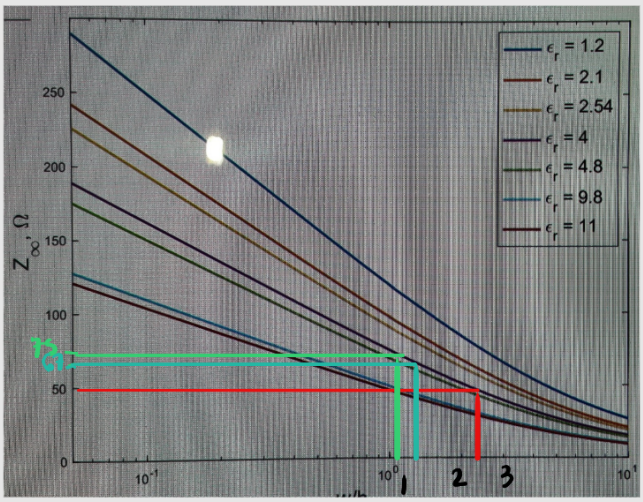
To ensure matching at port 1 the equivalent admittances must equal the reference admittance: $\Rightarrow$ reference impedance for all ports

$$\frac{Z_0}{Z_{\infty 2}^2} + \frac{Z_0}{Z_{\infty 3}^2} = \frac{1}{Z_0} \Rightarrow \frac{Z_0}{1{,}25\, Z_{\infty 3}^2} + \frac{Z_0}{Z_{\infty 3}} = \frac{1}{Z_0} \Rightarrow \frac{Z_0}{Z_{\infty 3}^2}\left(\frac{1}{1{,}25} + 1\right) = \frac{1}{Z_0} \Rightarrow$$

$$\Rightarrow \frac{Z_0}{Z_{\infty 3}^2}\, 1{,}8 = \frac{1}{Z_0} \Rightarrow Z_{\infty 3} = \sqrt{1{,}8\, Z_0^2} = \sqrt{1{,}8\; 50^2} = 67{,}08\,\Omega$$

Now using the initial expression:

$$Z_{\infty 2} = \sqrt{1{,}25\;\; 67{,}08^2} = 74{,}997\,\Omega \approx 75\,\Omega$$



- $67{,}08\,\Omega = Z_{\infty 3}$
- $75\,\Omega = Z_{\infty 2}$
- $50\,\Omega$

For the access line, for W/h we have to use the big formula with A and B, while on $Z_{\infty 2}$ and $Z_{\infty 3}$ we can use the other.

- For the access line:

$$A = \frac{50}{60}\sqrt{\frac{4{,}45+1}{2}} + \frac{4{,}45-1}{4{,}45+1}\left(0{,}23 + \frac{0{,}11}{4{,}45}\right) = 1{,}537$$

$$B = \frac{377\,\pi}{2 \cdot 50\sqrt{4{,}45}} = 5{,}615$$

$$\frac{W_A}{h} = \frac{2}{\pi}\left[5{,}615 - 1 - \ln(2 \cdot 5{,}615 - 1) + \frac{4{,}45-1}{2 \cdot 4{,}45}\left(\ln(5{,}615-1) + 0{,}39 - \frac{0{,}61}{4{,}45}\right)\right] = 1{,}897$$

$$\boxed{W_A = h \cdot 1{,}897 = 1{,}55 \cdot 1{,}897 = 2{,}94\ \text{mm}}$$

This is an access line, the length is arbitrary, the important one for us was just the width, in fact its width modifies the impedance.

• For $Z_{\infty 2}$

$$A = \frac{75}{60}\sqrt{\frac{5,45}{2}} + \frac{3,45}{5,45}\left(0,23 + \frac{0,11}{4,45}\right) = 2,225$$

$$\frac{W_{\infty 2}}{h} = \frac{8e^{A}}{e^{2A}-2} = \frac{8e^{2,225}}{e^{2\cdot 2,225}-2} = 0,885$$

$$W_{\infty 2} = 0,885 \cdot h = 0,885 \cdot 1,55 = 1,37 \text{ mm}$$

$$\varepsilon_{eff2} = \frac{5,45}{2} + \frac{3,45}{2}\frac{1}{\sqrt{1+\frac{12}{0,885}}} = 3,177$$

$$\lambda_{g2} = \frac{\lambda_0}{\sqrt{\varepsilon_{eff2}}} = \frac{51,72}{\sqrt{3,177}} = 29,0168 \text{ mm}$$

$$l_{\infty 2} = \frac{\lambda_{g2}}{4} = 7,254 \text{ mm}$$

$$\boxed{\begin{aligned} W_{\infty 2} &= 1,37 \text{ mm} \\ l_{\infty 2} &= 7,254 \text{ mm} \end{aligned}}$$

• For $Z_{\infty 3}$

$$A = \frac{67,08}{60}\sqrt{\frac{5,45}{2}} + \frac{3,45}{5,45}\left(0,23 + \frac{0,11}{4,45}\right) = 2,00679$$

$$\frac{W_{\infty 3}}{h} = \frac{8e^{2,00679}}{e^{2\cdot 2,00679}-2} = 1,116$$

$$W_{\infty 3} = 1,116 \cdot h = 1,116 \cdot 1,55 = 1,729 \text{ mm}$$

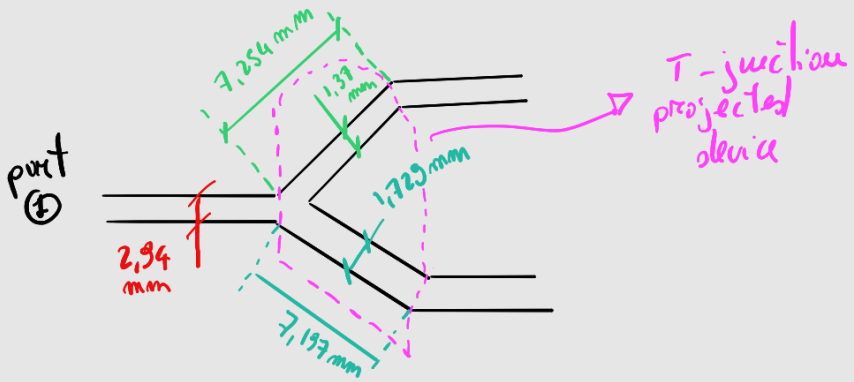
$$\boxed{\begin{aligned} W_{\infty 3} &= 1,729 \text{ mm} \\ l_{\infty 3} &= 7,197 \text{ mm} \end{aligned}}$$

$$\varepsilon_{eff3} = \frac{5,45}{2} + \frac{3,45}{2}\frac{1}{\sqrt{1+\frac{12}{1,116}}} = 3,228$$

$$\lambda_{g3} = \frac{\lambda_0}{\sqrt{\varepsilon_{eff3}}} = \frac{51,72}{\sqrt{3,228}} = 28,787 \text{ mm}$$

$$l_{\infty 3} = \frac{\lambda_{g3}}{4} = 7,197 \text{ mm}$$

The layout of the microstrip is:



MATLAB results for exercise 3

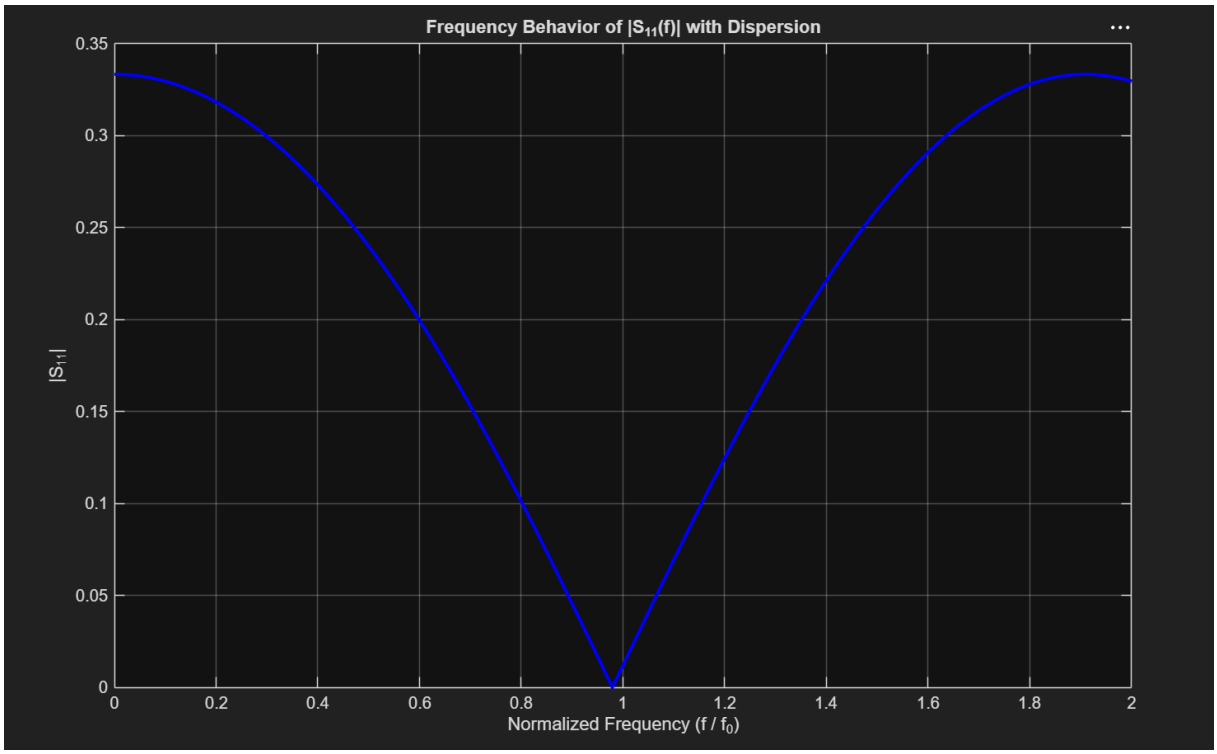


Figure 1: Frequency behavior of $|S_{11}(f)|$ taking dispertion into account

```
% ==================================================================
% Microstrip T-Junction Power Divider with Dispersion
% ==================================================================

% General Parameters
f0 = 5.8e9;
c = 3e8;
e_r = 4.45;
h = 1.55e-3;                    % Substrate thickness in meters
Z0 = 50;
Y0 = 1 / Z0;

% Z_inf2 in physical Parameters (Z_inf2 = 75 Ohm)
W2 = 1.37e-3;                   % Width in meters
l2 = 7.254e-3;                  % Physical length in meters
```

```
Y_inf2 = 1 / 75;

% Z_inf3 in pysical Parameters (Z_inf3 = 67.08 Ohm)
W3 = 1.729e-3;                % Width in meters
l3 = 7.197e-3;                % Physical length in meters
Y_inf3 = 1 / 67.08;

% Frequency Sweep Setup [0, 2*f0]
% Starting slightly above 0 (1 MHz) to avoid division by zero in dispersion formulas
f = linspace(1e6, 2*f0, 1000);
norm_f = f / f0;
lambda0 = c ./ f;

% Static Effective Permittivities
e_eff0_2 = (e_r + 1)/2 + ((e_r - 1)/2) * (1 / sqrt(1 + 12*h/W2));
e_eff0_3 = (e_r + 1)/2 + ((e_r - 1)/2) * (1 / sqrt(1 + 12*h/W3));

% Dispersive Effective Permittivities, this is frequency dependant!!
% F factors for Branch 2
F2 = 4 * (h ./ lambda0) * sqrt(e_r - 1) .* (0.5 + (1 + 0.868 * log(1 + W2/h)).^2);
% F factors for Branch 3
F3 = 4 * (h ./ lambda0) * sqrt(e_r - 1) .* (0.5 + (1 + 0.868 * log(1 + W3/h)).^2);

% e_eff(f) for branch 2
e_eff_f2 = (sqrt(e_eff0_2) + (sqrt(e_r) - sqrt(e_eff0_2)) ./ (1 + 4 * F2.^(-1.5))).^2;
% e_eff(f) for branch 3
e_eff_f3 = (sqrt(e_eff0_3) + (sqrt(e_r) - sqrt(e_eff0_3)) ./ (1 + 4 * F3.^(-1.5))).^2;

% Frequency-Dependent Electrical Lengths
theta2 = (2 * pi * f / c) .* sqrt(e_eff_f2) * l2;
theta3 = (2 * pi * f / c) .* sqrt(e_eff_f3) * l3;
```

```
% Input Admittances of the Two Branches
Yin2 = Y_inf2 .* ( (Y0 + 1i * Y_inf2 .* tan(theta2)) ./ (Y_inf2 + 1i * Y0 .* tan(theta2)) );
Yin3 = Y_inf3 .* ( (Y0 + 1i * Y_inf3 .* tan(theta3)) ./ (Y_inf3 + 1i * Y0 .* tan(theta3)) );

% Total Input Admittance & Reflection Coefficient at T-Junction
Yin_total = Yin2 + Yin3;
S11 = (Y0 - Yin_total) ./ (Y0 + Yin_total);

% 10. Plotting the Results
figure;
plot(norm_f, abs(S11), 'LineWidth', 2, 'Color', 'b');
title('Frequency Behavior of |S_{11}(f)| with Dispersion');
xlabel('Normalized Frequency (f / f_0)');
ylabel('|S_{11}|');
grid on;
```